

Correlation between penile duplex findings and stress electrocardiography in men with erectile dysfunction.



The aim of this work is to assess the association between vasculogenic erectile dysfunction (ED) and coronary artery disease in men above the age of 40 y. The study included 40 patients above 40 y of age with vasculogenic ED of more than 3 months duration. A dynamic duplex study after intracavernosal injection of a bimix solution (60 mg papaverine + 2 mg phentolamine mesylate) was carried out using a color ultrasound machine. The patients underwent a stress ECG test, carried out on a motor-driven treadmill according to the 'Bruce Protocol'. A total of 12 patients were diagnosed with positive ischemic heart disease (IHD). Their mean peak systolic velocity (PSV) was $PSV = 19.58$ cm/s. In all, patients were diagnosed with negative IHD; their mean PSV was 36.21 cm/s. A statistically significant difference was observed between patients with positive IHD and patients with negative IHD regarding PSV ($P = 0.003$). The sensitivity of a PSV of less than 35 cm/s in predicting IHD was 50% with a specificity of 100%. Positive predictive value for abnormal stress ECG to predict a PSV of less than 35 cm/s was 100%. In conclusion, the PSV of cavernosal arteries is a reliable measure for predicting IHD in patients with vasculogenic ED. Patients with a PSV of less than 35 cm/s should be referred for cardiologic assessment as they carry a real risk of having silent IHD.